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VHR-Seg: A Coarse-to-Fine Deep Learning Framework for Semantic Segmentation in Very High Resolution Remote Sensing Imagery

by

Yaowen Chang

Email: yc5224@imperial.ac.uk
GitHub username: esemsc-yc5224
Repository: <https://github.com/ese-ada-lovelace-2024/irp-yc5224>

Supervisors:

Dr. Yves Plancherel
Ms. Myriam Prasow-Emond

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Abstract

Semantic segmentation of very high resolution (VHR) remote sensing imagery (RSI) holds great potential for applications in ecological conservation, urban management, and many other domains. However, the large scale of VHR imagery, the high annotation cost, and the lack of standardized workflows present major challenges for VHR RSI semantic segmentation. These challenges make it difficult to develop generalizable models and limit the scalability of existing workflows, resulting in most existing deep learning approaches being restricted to specific datasets, which limits their cross-domain applicability.

This study addresses these challenges by proposing VHR-Seg, a scalable two-stage framework for coarse-to-fine semantic segmentation of VHR RSIs. The framework provides end-to-end functionalities, including automated geospatial data processing, model training and inference, and annotation guidance, where segmentation can be performed according to user-defined regions of interest. To achieve more accurate fine-grained segmentation, the High-Resolution (HR) series models are designed to capture fine details while preserving contextual semantic information.

To evaluate the performance of VHR-Seg, comparative experiments were conducted between HR-series models, baseline approaches, and state-of-the-art (SOTA) methods under the same experimental settings. The results demonstrate that the proposed models are both effective and efficient. Exploratory discussions on image resolution and training dataset scale reveal the varying requirements of fine-grained segmentation across different categories. The potential of leveraging the distinct characteristics of each stage in VHR-Seg to enhance annotation efficiency and accuracy has also been investigated.

1 Introduction

Satellite and airborne remote sensing is currently undergoing a technical revolution with the appearance and blooming development of very high-resolution sensors. Benefiting from advances in sensor technologies, optical systems, and data processing algorithms, the spatial resolution of imagery from satellites and unmanned aerial vehicles (UAVs) has improved dramatically in recent years [10], reaching sub-meter levels for commercial satellites [20, 36] and centimeter levels for UAVs [28, 30]. Improvements in the spatial resolution of VHR RSIs have substantially broadened their application potential. For example, researchers apply VHR remote sensing images (RSIs) to map features as small as insects and other fauna with centimeter-level precision over vast areas in ecological studies. These breakthrough achievements make high-frequency biodiversity monitoring possible and provide critical data for tracking endangered species and their recovery plans. In addition to ecological applications, VHR urban remote sensing data can be used to generate accurate urban functional area distribution maps [21], which is of great significance to improve urban governance capacity and achieve the sustainable development goals [37].

The huge application potential of VHR RSIs has further intensified the demand for their utilization. The growing demands to utilize VHR RSIs call for new data processing methods [44]. In recent years, a large number of researchers have adopted machine learning (ML) and deep learning (DL) methods to analyze RSIs [1]. Among the various tasks, semantic segmentation has emerged as one of the most crucial, as it focuses on the precise delineation of semantic regions within remote sensing imagery [5, 42]. The advent of deep learning has greatly promoted this field and led to the development of numerous effective algorithms.

Early research focused on designing CNN-based architectures [43] to capture local features and hierarchical representations. However, CNN-based models remained constrained in model-

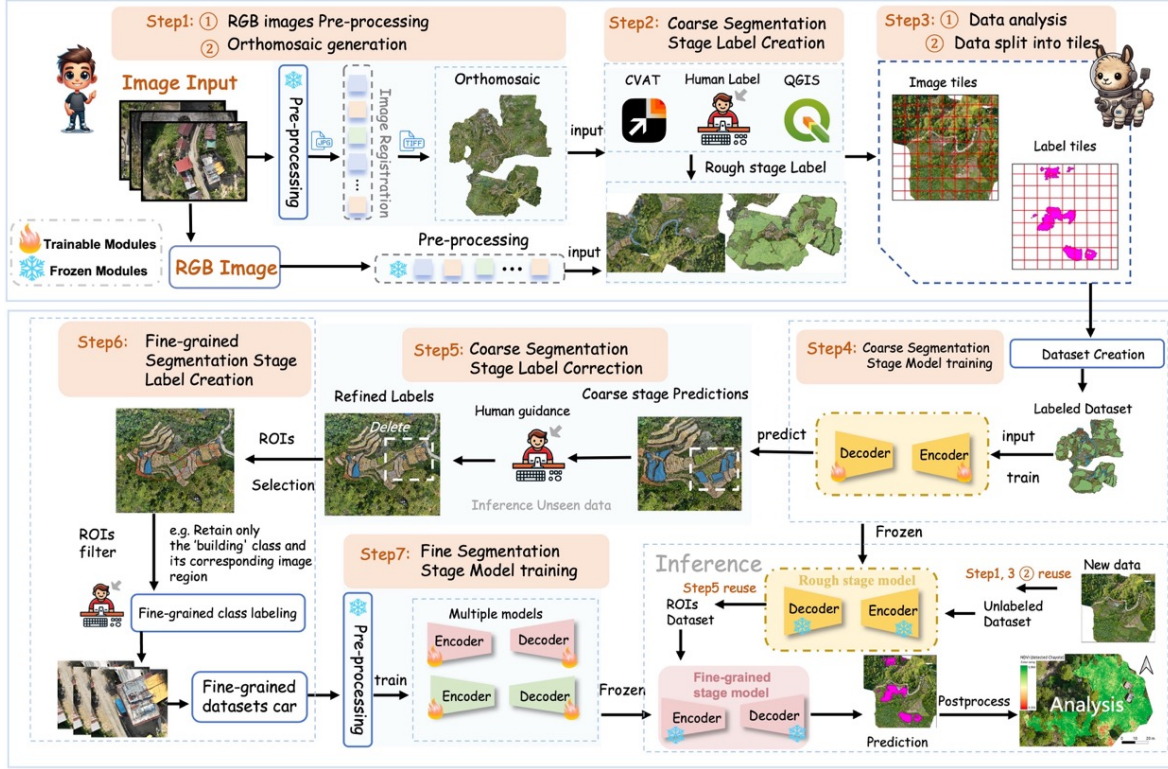


Figure 1: Overall framework of VHR-Seg. The framework consists of seven main steps. Model training, data preprocessing, and data postprocessing are automated, while label creation, correction, and ROI selection rely on human expertise.

ing global dependencies, which has motivated the adaptation of Transformer architectures [38] for semantic segmentation of RSIs. An example of such adaptation is UNetFormer [39], which proposes an UNet-like Transformer (UNetFormer) for real-time remote sensing segmentation. Recently, foundation models have been developed, aiming to achieve generalization across diverse domains without task-specific retraining [11]. As a representative foundation model in semantic segmentation, the Segment Anything Model (SAM) [24] is designed for promptable segmentation across arbitrary scenes without additional tuning. Building upon SAM, its successor SAM2 [34] extends these capabilities to instance-level segmentation in both image and video tasks. Designed to cover as many use cases as possible, SAM2 is a robust solution for segmenting commonly occurring features. However, despite their strong generalization in natural images, foundation models such as SAM and SAM2 still struggle in VHR RSIs segmentation due to their unique characteristics. Application of SAM2 to this dataset is shown in Figure A.1. Although many semantic segmentation methods have achieved great success in the domain of RSIs, they often fail to achieve comparable performance when directly migrated to VHR RSIs. There exist three challenges in semantic segmentation in VHR RSIs.

Spatial size difference. VHR RSIs are usually much larger in size than typical natural images used in computer vision applications. For example, images in the PASCAL VOC data set [18] (an open benchmark for semantic segmentation of natural images) have a size of 500×375 pixels, whereas images in the ISPRS semantic labeling data set (Potsdam area) [16] have 6000×6000 pixels and the original images in the Beijing land-use classification data set have 47244×47244 pixels. The significant difference in the spatial size makes it impossible to give a whole VHR RSI as input to semantic segmentation models; otherwise, the computational costs would be unaffordable [26].

Lack of standardization and reproducibility workflows. Current remote sensing (RS) re-

search in VHR mapping faces a major challenge due to the inconsistency in deep learning (DL) models design and usage. Many studies develop custom models tailored to specific datasets or features [1, 29], and literature reviews have revealed a notable lack of standardized DL workflows [19, 23, 31]. This hinders the transferability of DL implementations, especially since VHR RS datasets are diverse and feature rich. It becomes infeasible to build new models and workflows from scratch for each new feature studied.

Scarcity of annotated data poses a major obstacle to the advancement of VHR RSIs semantic segmentation. Due to the extremely high cost and time required for manual labeling at very fine spatial resolutions, large-scale annotated datasets are rare, which limits the training and generalization ability of deep learning models.

These challenges underline the need for practical and scalable frameworks for VHR RSIs segmentation. To address these challenges, a coarse-to-fine deep learning framework for semantic segmentation in very high resolution remote sensing imagery is proposed. The overall framework is illustrated in Figure 1. The objectives of this research are listed as follows: (i) design a scalable framework that integrates annotation, preprocessing, training, and inference into a reproducible pipeline; (ii) reduce computational and annotation costs by regions of interest (ROIs) selection in fine-grained segmentation stage; (iii) develop HR-series models capable of preserving contextual semantics while accurately segmenting fine-scale objects; and (iv) conduct systematic experiments on a real-world UAV-based datasets to evaluate the effects of image resolution and dataset scale on segmentation performance. Through these objectives, the study aims to provide a practical, efficient, and reproducible solution for semantic segmentation of VHR remote sensing imagery.

2 Methodology

2.1 Study Area and Data

The dataset to be used in this project is a series of 5280 x 3956 RGB images and an RGB orthomosaic formed from these UAV images. The imagery covers over 2 km² of the legacy mining site of Sto. Niño in Tublay, Benguet Province, Philippines. The overview of the dataset is provided in the Appendix B. A DJI Mavic 3 MS drone equipped with a Zenmuse L1 lidar/photo camera was flown at 50m altitude to capture this data in November 2023. The spatial resolution ranges from 2 to 3 cm/px.

2.2 Dataset Annotation

A two-stage coarse-to-fine annotation strategy was employed to balance annotation efficiency with segmentation quality. The dataset was annotated using QGIS and CVAT. The workflow is shown in Figure 2.

This two-stage process allows both broad coarse-level annotations and subsequent fine-grained annotations to be incorporated seamlessly. In the first stage, a subset of the imagery was annotated into coarse land-cover categories, including forest, cropland, grassland, building, road, and bare soil. These labels could be generated relatively quickly because annotators only needed to distinguish major landscape types rather than object-level boundaries. A preliminary coarse segmentation model was then trained and applied across the entire dataset to generate semantic masks. Based on these masks, users can select regions of interest (ROIs). For example, if palm trees are the research target, annotation will be restricted to forest areas.

In the second stage, fine-grained labels were created only within the selected ROIs—for instance, palm trees and banana trees were annotated within forests, while irrelevant areas were

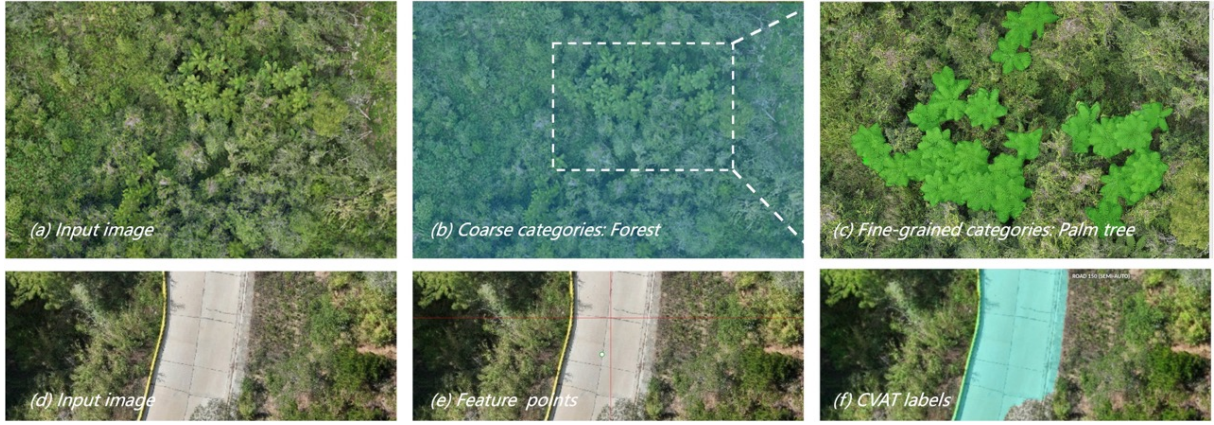


Figure 2: Labeling workflow for the Sto. Niño dataset. First row: QGIS-based annotation. (a) Palm plantation region; (b) coarse category label "forest" with blue mask; (c) refinement to fine-grained category "palm tree" (green). Second row: CVAT-based annotation. (d) Input image; (e) user-placed points indicating target objects; (f) automatically generated labels.

excluded. By restricting fine annotation to these ROIs, the amount of detailed labeling required on VHR RSIs was significantly reduced.

This hierarchical workflow reduced annotation workload while allowing flexible and incremental extension when new areas or target classes were introduced. To ensure reproducibility, documentation describing the QGIS labeling procedure is provided in Appendix C. In this project, forest, road, and building were selected as coarse categories, whereas palm tree, banana tree, and car (Appendix D) served as fine-grained categories for segmentation.

2.3 VHR-Seg: An Open-Source Pipeline for Coarse-to-Fine Segmentation

VHR-Seg is a platform-independent command-line tool implemented in Python 3.10. The system is modular and designed for practical geospatial deployments. The repository includes step-by-step setup and usage notes, and the documentation is auto-generated to HTML with Sphinx. To ensure reproducibility across machines, all dependencies were managed using `pip-tools`, which compiles a reproducible `requirements.txt` file from a lightweight specification. Therefore, users can install the full environment on new machines with a single command. To avoid complications with GDAL's OS-specific dependencies, Anaconda was used to manage the library's integration. VHR-Seg's deep learning modules are implemented with the MMSegmentation framework, enabling flexible configuration for coarse-grained and fine-grained models. All components rely exclusively on open-source software.

2.3.1 System Architecture

As shown in Figure 1, the VHR-Seg pipeline consists of seven stages, which can be grouped into three main blocks: preprocessing, training and evaluation, and the postprocessing of predictions into final segmentation outputs. To design the code, VHR-Seg uses 'workflow' to represent these steps. A UML diagram illustrating the VHR-Seg workflows is provided in the Appendix E, where the interconnections between different workflows are described in detail. For clarity, we also refer to a workflow as a "task". VHR-Seg supports both the execution of individual tasks and running the entire pipeline in a single process. To ensure efficiency, a multithreaded design pattern has been adopted. Further implementation details are discussed in the Appendix F. Through the command-line interface (CLI), users can choose to run only

specific parts of the pipeline, instead of the whole process. They can also update just the fine-grained stage whenever new regions of interest (ROIs) or new subclasses are added, without having to re-run everything from the beginning. In addition, multiprocessing is supported to accelerate large-scale inference over thousands of tiles, and the system works with common geospatial raster formats (GeoTIFF, PNG, JP2, etc.) and polygon label files (Shapefiles, COCO JSON, CVAT XML). A complete list of available tasks and their dependencies is provided in Appendix F.

2.3.2 Data Preprocessing

In VHR-Seg workflows, two types of input imagery are typically used: raw UAV images prior to orthorectification, and georeferenced orthophotos or other large-scale geospatial products. While deep learning models usually process relatively small patches (e.g., 512×512 pixels), geospatial datasets can be orders of magnitude larger. A common solution is downsampling, but in VHR RSI semantic segmentation the main advantage lies in preserving fine details, making aggressive downsampling unsuitable. Splitting large RS imagery into tiles is an alternative method to deal with this situation. However, this method also presents challenges: orthomosaics often exceed GPU memory limits, and tiling may introduce boundary artifacts because adjacent patches are processed independently. VHR-Seg addresses these issues by processing input images through memory mapping, avoiding the need to load entire files into memory.

To maintain overlap across adjacent tiles, a sliding-window strategy is adopted. This improves prediction consistency and produces roughly $(1/s)^2$ times more training samples compared to non-overlapping tiling, where $0 < s < 1$ denotes the stride ($s = 0.5$ in our experiments). Output image and label tiles are saved as compressed JPG (for images) or PNG (lossless, for labels). Georeferencing from the original input is preserved and reapplied, ensuring that the final segmentation maps remain correctly aligned when reimported into GIS software.

2.3.3 Utilizing MMSegmentation

Recently, an increasing number of well-maintained open source projects [4, 8, 14] have used MMSegmentation as a foundation framework for developing custom models and tools. The MMSegmentation framework offers several advantages: it is actively maintained, supports a wide range of state-of-the-art (SOTA) semantic segmentation methods [13], and provides both high efficiency and strong extensibility. The core of MMSegmentation lies in highly abstracted, simple configuration files that automatically generate PyTorch models. Compared to directly using PyTorch or other deep learning libraries, using the MMSegmentation framework has proven to be both effective and sustainable. Given these advantages, VHR-Seg adopts MMSegmentation as the backbone framework, providing flexible methods for model training and inference within the hierarchical segmentation pipeline.

Training methods. Semantic segmentation datasets often suffer from severe class imbalance, a problem also observed in the Sto. Niño dataset. In order to mitigate the imbalance between majority classes and minority classes, a class-balance algorithm was proposed, as shown in Appendix G.1. The weights will be assigned to every image in the training set, which is a kind of oversampling. Although oversampling may increase the model performance, it may introduce redundancy, causing model overfitting. To deal with the potential problem, randomized image augmentations are applied by using Albumentations library [3]. Augmentations like downscaling and color shifts increase tile diversity on the training set [25, 35]. The more diverse the tiles, the less likely the model is to overfit, thereby improving its generalization ability.

Dataset splitting. VHR-Seg allows users to create customized datasets by selecting their

regions of interest (ROIs). Besides, it also provides an automatic mode that randomly splits the dataset. However, remote sensing data are typically spatially auto-correlated [11]. Therefore, dividing datasets based on users' interest can help reduce spatial leakage and mitigate model overfitting.

2.3.4 Postprocessing

Since the model operates on partially overlapping tiles, pixel-wise predictions must be merged back into a seamless full-resolution segmentation map. The merging strategies have a direct impact on segmentation map visual quality and consistency. Two merging strategies are supported in this pipeline: (1) logit-merge, which fuses overlapping regions by weighting them with the model's confidence scores. (2) crop-merge, which discards boundary areas of tiles and keeps only their centers, except near image edges. The crop-based approach generally produces more accurate results while also being computationally faster. Detailed information is provided in Appendix G

2.4 Model Design and Selection

As discussed in Section 1, VHR semantic segmentation typically requires substantial computational resources. However, existing pre-trained models [7, 41] often perform poorly in fine-grained segmentation tasks involving highly similar categories, such as palm trees and banana trees. This motivates the need to improve fine-grained segmentation performance while preserving the benefits of VHR imagery under controllable computational costs. To this end, SegFormer and Mask2Former, both of which have demonstrated state-of-the-art (SOTA) performance [11] on natural image benchmarks, are adopted as backbone networks. Inspired by HRDA [22], these backbones are combined with a dual-resolution encoder-decoder head, resulting in the proposed HRsegformer and HRmask2former. These models achieve superior performance on fine-grained targets. The detailed model design is provided in Appendix J.

To comprehensively evaluate the performance of our proposed models, UNet [15] was selected as the baseline model. The comparison results are presented in Section 3.3. In addition, an ablation study was conducted to validate the effectiveness of the HR-series models and to isolate the contribution of the dual-resolution design, as reported in Appendix K.

2.5 Experiment

2.5.1 Experiment Design

A comprehensive evaluation of VHR-Seg considered three key survey factors: object scale, Ground Sample Distance (GSD), and the amount of training data. To examine the effect of object scale, models were trained separately on datasets from the coarse and fine-grained stages of our framework, which capture objects of markedly different sizes.

To assess sensitivity to spatial resolution, models were trained for each feature type on twelve dataset versions with progressively coarser GSDs, nine of which correspond to common VHR satellite sensors (Appendix H). The upper limit was set to 0.08 m/px due to computational constraints. GSD degradation was performed using several synthetic downsampling strategies (Figure 3), as detailed in Appendix I. Among them, method B degraded only the image resolution while keeping annotations intact. Considering multiple degradation schemes provides a more reliable view of performance trends and reduces bias from any single method.

Finally, the influence of training set size was analyzed by progressively reducing the number of training images. In total, 120 models were trained across these experimental settings.

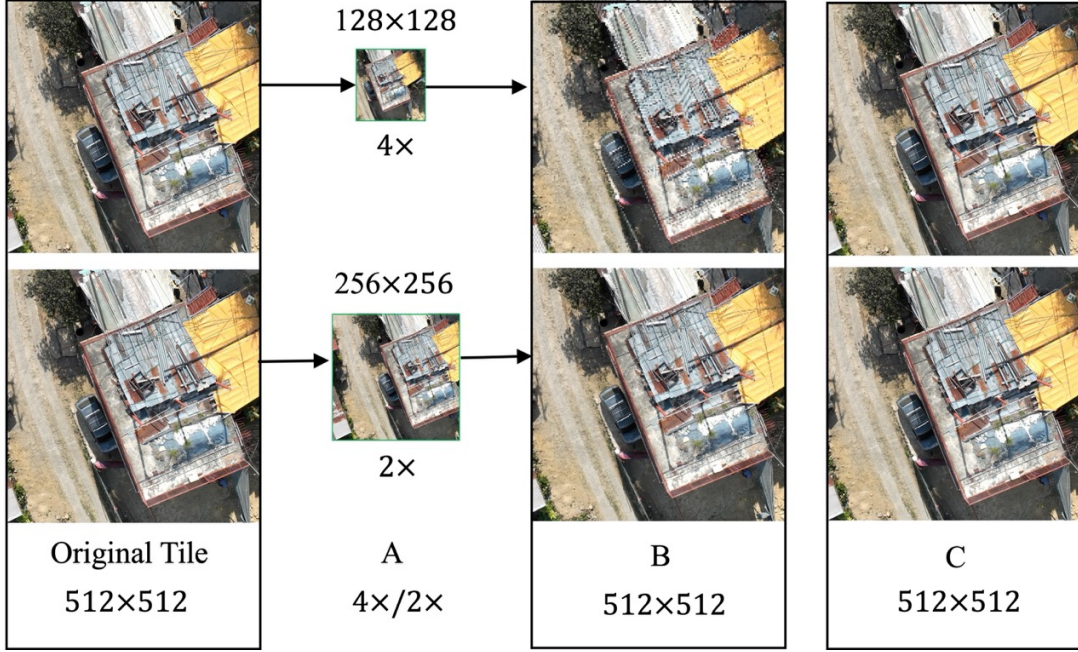


Figure 3: Different approaches for reducing the resolution of orthomosaic tiles (A–C) and their impact on upsampling recovery

2.5.2 Experiment Setup

Models were trained on the high-performance computing (HPC) platform of Imperial College London in both stages, using a single NVIDIA A100 GPU (40 GB VRAM), 32 GB of system memory, and 4 Intel CPU cores. To support reproducibility, the same random seed was applied across all libraries, with consistent software versions maintained (CUDA 12.1, PyTorch 2.1.0+cu121, and TorchVision 0.16.0+cu121). Although exact bitwise reproducibility across hardware cannot be guaranteed, these settings help to minimize variability. A fixed batch size and random seed ensure that training samples are presented in a consistent order across runs, regardless of the hardware environment. To reduce spatial leakage and maintain balanced class distributions, the orthomosaic imagery and corresponding labels were manually split into 70% training, 10% validation, and 20% test subsets (Figure 4). The same ratio was applied in the fine-grained stage for the selected ROIs. After dataset preparation, we trained different models for the coarse and fine-grained stages. Mask2Former [7] was used as the coarse-stage segmentation model. The model was configured with a Swin-S backbone [17] pretrained on the Cityscapes dataset [12]. The backbone weights were unfrozen during training to allow fine-tuning and improve transfer learning capacity. In the fine-grained stage, HRsegformer and HRmask2former were employed to segment the fine-grained classes, using SegFormer-B2 and Swin-S backbones, respectively.

For all stages, models were trained with an initial learning rate of 6×10^{-5} , which was scheduled by the poly strategy with power 0.9 over 200 epochs. The optimizer was AdamW [27], with $\epsilon = 10^{-8}$, weight decay 10^{-4} , and a batch size of 4 on each GPU. No hyperparameter tuning was required in the coarse stage. In the fine-grained stage, we analyzed the downsampling factor s and the relative crop size of the HR-series models. The ablation study in Appendix K showed that the models were robust with respect to s , with 2 was chosen as a suitable value.

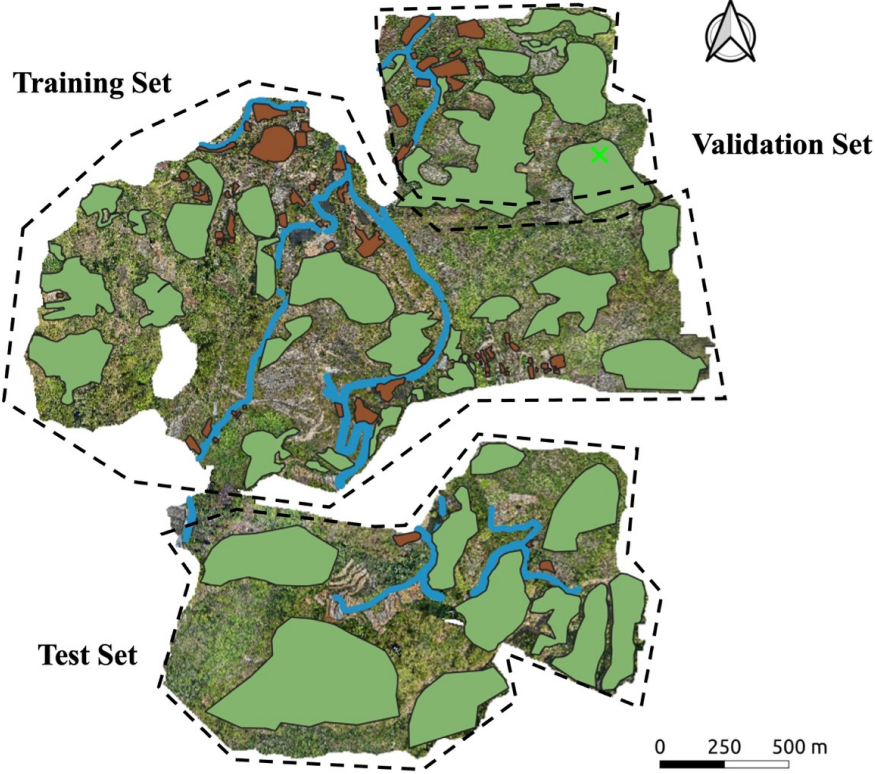


Figure 4: Dataset split of the Sto. Niño imagery into training, validation, and test regions in coarse stage. Ground truth annotations are shown, containing a balanced representation of classes. A buffer zone was added to separate training from evaluation regions, reducing spatial leakage.

2.5.3 Evaluation Metric

Following standard evaluation protocols from prior work [9, 40], intersection over union (IoU), Dice coefficient, mean intersection over union (mIoU), and mean Dice coefficient (mDice) were used to evaluate model performance. IoU measures segmentation accuracy by comparing the overlap area with the union of predictions and ground truth [11]. The Dice coefficient is the harmonic mean of precision and recall, where precision denotes the fraction of positive predictions that are correct, and recall denotes the proportion of positive samples correctly identified by the model. The mIoU is computed as the average IoU across all classes:

$$\text{IoU}_i = \frac{TP_i}{TP_i + FP_i + FN_i}, \quad \text{mIoU} = \frac{1}{N} \sum_{i=1}^N \text{IoU}_i, \quad (1)$$

The Dice coefficient and the mean Dice coefficient (mDice) are defined as:

$$\text{Dice}_i = \frac{2TP_i}{2TP_i + FP_i + FN_i}, \quad \text{mDice} = \frac{1}{N} \sum_{i=1}^N \text{Dice}_i, \quad (2)$$

where TP_i , FP_i , and FN_i represent true positives, false positives, and false negatives for class i , and N is the total number of classes. In addition, the complexity of the model and the inference speed were measured in terms of FLOPs and FPS. Memory usage was measured by the number of parameters (P) and the overall memory footprint.

3 Results

3.1 Profiling of VHR-Seg Performance

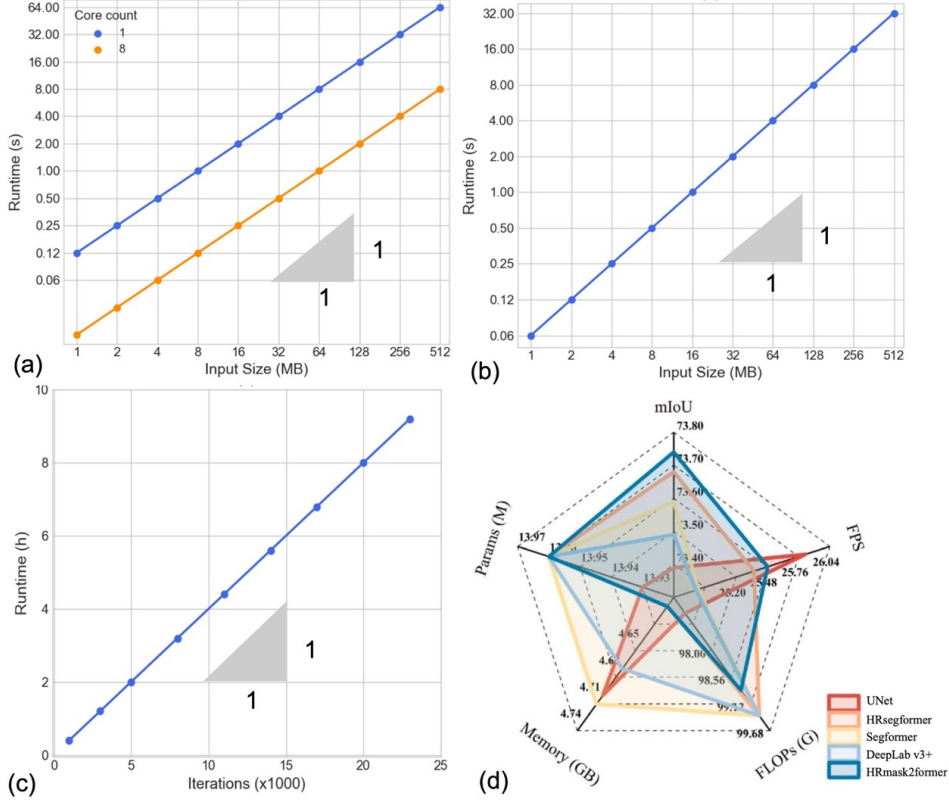


Figure 5: VHR-Seg performance in different tasks. (a) Image dataset splitting performance; (b) postprocessing runtime; (c) iteration runtime. (d) Comprehensive comparison of performance and efficiency for different segmentation models used in VHR-Seg

To evaluate the computational efficiency of VHR-Seg, both data processing methods and model training routines were profiled, as shown in Figure 5. Since image splitting and tile merging both iterate once over the $n \times m$ input, their theoretical complexity is $\mathcal{O}(nm)$. With GPU acceleration, the training stage scales proportionally with input size.

In practice, runtime variations were observed and needed to be addressed. When VHR-Seg divides images with a stride of 0.5, the workload per image quadruples. This overhead is alleviated by parallelizing row-wise operations across CPU cores. Although tile merging also follows $\mathcal{O}(n)$ complexity, the crop-merge strategy significantly speeds up execution by avoiding redundant boundary computations and leveraging vectorized NumPy implementations.

In addition, a comprehensive comparison of accuracy and efficiency for different segmentation models used in VHR-Seg is shown in Figure 5 (d). The radar chart compares models along five dimensions: mIoU, inference speed Frames Per Second (FPS), FLOPs, memory usage, and parameter count. The results show that the proposed HRsegformer and HRmask2former achieve higher accuracy while maintaining competitive efficiency, demonstrating a favorable trade-off between accuracy and computational cost.

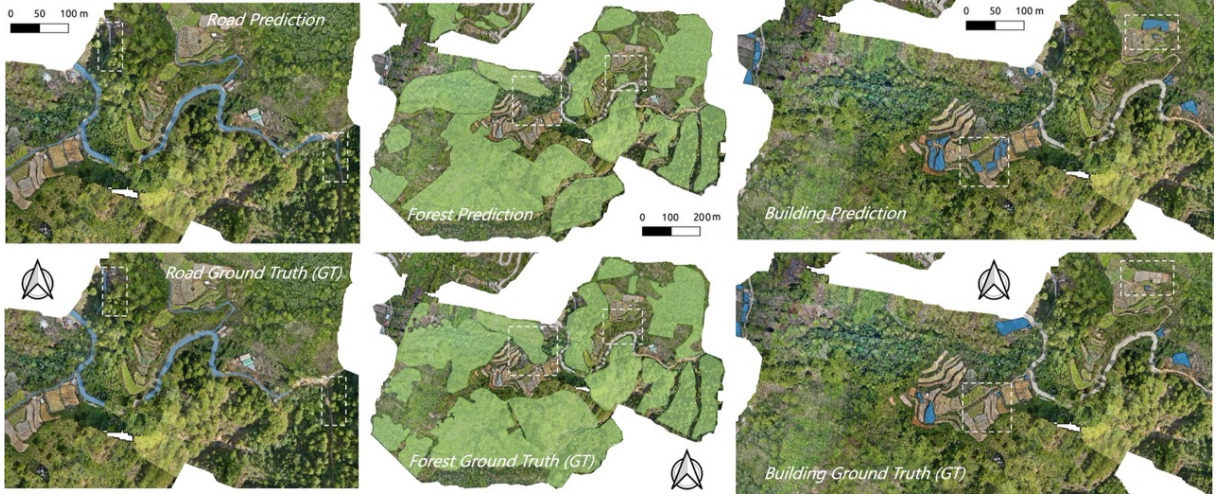


Figure 6: Coarse-stage semantic segmentation results. Predicted maps (top row) are compared with ground truth annotations (bottom row) in the test set. Forest, road, and building are indicated in green, cyan, and blue, respectively.

3.2 Coarse Stage Semantic Segmentation Performance Using VHR-Seg

Firstly, we undertake a comprehensive evaluation of the proposed VHR-Seg within the coarse stage, including both qualitative and quantitative analysis. As illustrated in Figure 6, the coarse-stage segmentation results show a strong alignment with the ground truth labels across all three categories.

The model effectively identifies large-scale land features, and its predictions capture the dominant landscape structures in a coherent way. Forest regions, in particular, are segmented with high accuracy. The more fragmented patterns of roads and buildings are also identified with good precision. These outcomes are confirmed by the mIoU scores, which were 80.35 (forest), 74.28 (road), and 71.32 (building).

Methodologically, these coarse predictions are important because they enable the targeted selection of regions of interest (ROIs) for the subsequent fine-grained analysis. By constraining annotation and modeling efforts to these ROIs, the framework achieves substantial reductions in computational and labeling costs while retaining the advantages of VHR RSIs. These findings demonstrate that the coarse stage establishes a robust and cost-effective foundation for the hierarchical segmentation workflow proposed in this study.

3.3 Fine-grained Stage Semantic Segmentation Performance Using VHR-Seg

In this section, we compare VHR-Seg with previous SOTA approaches in fine-grained stage. In this stage, both **HRsegformer** and **HRmask2former** trained with VHR-Seg achieved better performance than baseline models, particularly on small object categories. As shown in Table 1 and Table 2, the proposed models obtained improvements in Palm Tree (best 55.73 in IoU) and Banana Tree (best 60.25 in IoU).

These improvements are further supported by the qualitative results in Figure 7, where the proposed models capture tree crowns more completely and reduce fragmented predictions compared to U-Net [4] and DeepLab v3+ [6]. For the Car category, which occupies smaller regions in the VHR images, HRsegformer demonstrates competitive performance (75.98 Dice, 65.26 IoU), successfully distinguishing vehicles from surrounding objects. Qualitative examples



Figure 7: Visualization of fine-grained segmentation results on the test dataset. Green indicates forest, light blue indicates roads, and blue indicates buildings. Predicted results (top row) are generated by the proposed model, while the ground truth (bottom row) corresponds to the manual annotations.

Table 1: Experimental results on fine-grained segmentation stage The **bold** denotes the best performance among different methods. Δ is boost over U-Net baseline. The evaluation metric is the Dice and mDice.

Method	C-to-F	mDice	Palm Tree	Banana Tree	Road	Car	Building	Δ
U-Net [4]	✗	63.99	48.13	56.76	71.21	64.72	79.12	-
Segformer w/ b2 [41]	✗	66.56	53.67	58.73	74.45	68.49	77.51	-
DeepLab v3+ [6]	✗	64.81	51.69	54.72	72.96	65.79	78.89	-
Mask2former [7]	✗	70.19	59.39	63.89	77.18	70.29	80.23	-
HRsegformer w/ b0	✓	76.04	68.57	72.19	<u>80.33</u>	75.98	83.13	+12.05
HRmask2former	✓	<u>75.86</u>	<u>65.64</u>	<u>70.71</u>	82.24	<u>74.61</u>	86.10	+11.87

Table 2: Experimental results on fine-grained segmentation stage The **bold** denotes the best performance among different methods. Δ is boost over U-Net baseline. The evaluation metric is the IoU and mIoU.

Method	C-to-F	mIoU	Palm Tree	Banana Tree	Road	Car	Building	Δ
U-Net [4]	✗	51.38	35.26	43.54	60.28	52.35	65.46	-
Segformer w/ b0 [41]	✗	54.28	40.52	44.65	63.20	55.63	67.38	-
DeepLab v3+ [6]	✗	52.25	37.64	40.57	61.25	52.43	69.34	-
Mask2former [7]	✗	58.33	45.34	50.26	66.92	57.84	71.28	-
HRsegformer w/ b2	✓	<u>65.66</u>	55.73	60.25	71.43	65.26	75.64	<u>+14.18</u>
HRmask2former	✓	65.80	<u>52.56</u>	<u>58.37</u>	74.28	<u>63.42</u>	80.35	+14.42

highlight that cars are more consistently identified by the proposed models, whereas baseline methods often miss or merge them into the background. Both qualitative and quantitative results demonstrate that HRsegformer and HRmask2former are capable of handling fine-grained vegetation and man-made classes.

4 Discussion

4.1 VHR-Seg Models Performance

In fine-grained stage, the proposed HRsegformer and HRmask2former models achieve notable improvements over baseline models. Benefiting from their specially designed architecture that integrates both High-Resolution (HR) and Low-Resolution (LR) branches, they can distinguish small objects accurately. The LR branch provides global contextual information, while the HR branch captures details information. This dual path ensures that semantic consistency is maintained while preserving detailed object boundaries. Thus, HR-series models are capable of getting a better segmentation performance. The ablation study is in Appendix K.

However, while the HR-series models improve segmentation accuracy, several limitations remain. For example, the dual-resolution architecture comes with increased computational costs compared to lightweight CNN-based models. Although the efficiency–accuracy trade-off is favorable, large-scale deployment (e.g., over tens of thousands of UAV images) still requires substantial GPU resources, which may limit applicability in resource-constrained environments. And the performance gain is most pronounced for classes with sufficient training data and clear structural features. Minority or highly fragmented classes can not be detected accurately.

Future research may explore strategies such as domain adaptation to transfer knowledge from related domains into the VHR RSIs setting. Combining domain adaptation with semi-supervised or active learning could further reduce dependence on large annotated datasets and improve model robustness across diverse geospatial environments.

4.2 Effects of on Survey Factors Per-Class Model Performance

4.2.1 Relationship between GSD and DL Performance

As described in Section 2.5.1, we examined three key survey factors: object size, Ground Sample Distance (GSD), and training data quantity. Figure M.1 and Figure M.2 show model performance under varying conditions, with detailed GSD degradation scores provided in Appendix L. The variation in survey extent with spatial resolution is illustrated in Figure N.1.

The two-stage segmentation results indicate that fine-grained categories are sensitive to spatial resolution. Because these classes usually occupy fewer pixels in imagery, they require higher spatial resolution (i.e., smaller GSD values) to preserve sufficient detail for accurate discrimination. In contrast, coarse-level categories generally cover larger areas with richer pixel information, making them more robust to resolution degradation. From the results in Appendix L, it is evident that classifying coarse classes is much easier than classifying fine classes at a GSD of 0.5 m/px or higher. The performance drop in fine-grained classes is steeper than that of coarse classes as GSD increases. This phenomenon reveals that fine-grained classes are more strongly dependent on clear boundaries and texture information.

Interestingly, even within coarse-level categories, the effect of resolution is not same. For example, once GSD exceeds 0.5 m/px, performance decreases sharply in forest class, which may be due to its highly fragmented boundaries and spectral similarity to other vegetation types. However, road and building predictions remain stable in this process, as their structures exhibit clearer geometric patterns that are less affected by moderate resolution degradation.

These results suggest that the choice of GSD should be guided by the target classes of interest in practical applications. Selecting an appropriate resolution can help achieve an effective trade-off between acquisition cost and model accuracy.

4.2.2 Tradeoffs between Dataset Size and DL Performance

Dataset size is also an important factor that must be considered. As shown in Figure M.1 and Figure M.2, both coarse-level categories and fine-grained vegetation classes exhibit a similar trend: model accuracy improves rapidly as the training set increases, but the gain becomes marginal once the dataset exceeds roughly 30% of the total area. This suggests that, in our case study, including about 30% of the area in the training set may already provide stable model performance by ensuring adequate generalization to unseen data.

This conclusion, however, warrants cautious interpretation. The observed threshold may vary under different ecological or urban contexts. In particular, rare or highly fragmented classes may still require larger training sets to capture sufficient semantic information and avoid under-fitting.

Therefore, choosing an appropriate dataset scale is critical to balance data quantity with diversity when planning large-scale surveys. More broadly, carefully designed dataset sizing strategies not only improve model robustness but also ensure that computational resources are used efficiently, which is essential for scaling up deep learning applications in real-world remote sensing tasks.

4.2.3 Relationship between Object Size and Model Performance

Designed to capture fine-grained details, the HR-series model is a powerful architecture designed to capture fine-grained details in imagery. Combining global attention with scaled attention, it can effectively segment fine-grained classes inside ROIs. The HR encoder is able to extract detailed information such as textures, and the LR encoder captures contextual features surrounding the fine-grained objects. The LR branch usually acts as a teacher, providing guidance to the HR branch for more accurate segmentation.

Nevertheless, the performance of the HR-series models is fundamentally constrained by object size. When target objects occupy only a few pixels, as in imagery degraded to a GSD of 10 m/px, even the most sophisticated architecture cannot reliably identify them (Appendix L). This shows a key limitation of deep learning in remote sensing: models cannot fully compensate for low spatial resolution and tiny objects. Small, fragmented, or spectrally similar classes are still very difficult to detect.

For VHR RSIs segmentation, maintaining adequate GSD is as essential as choosing the right model, particularly for small or fine-grained targets.

4.3 VHR-Seg is an Efficient Labeling Pipeline

In our two-stage coarse-to-fine pipeline, an interesting phenomenon was observed in Figure 8 where the model occasionally predicted regions that were not annotated as ground truth, but were semantically correct in reality. These inconsistencies come from various factors, including annotator oversight, fatigue, or limited domain expertise. This observation reveals a common limitation of human-annotated datasets in remote sensing field: even when extensive effort is invested, labeling errors or omissions are difficult to avoid, and these can negatively influence downstream model training. In particular, if such mislabeled regions are directly treated as ground truth, they may introduce misleading training signals that bias the fine segmentation stage and degrade model performance.

The design of VHR-Seg can help solve this problem. The two-stage architecture allows users to conduct a coarse segmentation at first. After coarse segmentation, the ROIs selection allows users to further check the prediction maps, which provides an opportunity to detect and



Figure 8: Close-up view showing differences between ground truth road labels (white outlines) and model predictions (green). (1) Both ground truth and predictions were incorrect. (2) Ground truth was incorrect, but the model prediction was correct. (3) The model predicted a road region that was missing from the ground truth.

correct potential annotation errors before fine-grained segmentation, thereby preventing the propagation of wrong labels. So in the fine segmentation stage, the ROI-based strategy enables the model to concentrate on subtle intra-class differences without being confounded by inter-class variability. As a result, the model benefits from both a cleaner training signal and a more focused learning objective, improving its performance.

More importantly, cases such as (2) and (3) in Figure 8 demonstrate that models trained within VHR-Seg can assist in refining or correcting annotations, effectively turning the framework into an interactive labeling pipeline. In this way, VHR-Seg enhances both the efficiency of dataset construction and the robustness of fine-grained semantic segmentation in VHR remote sensing imagery.

5 Conclusion

This project proposed VHR-Seg, a coarse-to-fine deep learning framework for semantic segmentation of VHR RSIs. With its two-stage structure, VHR-Seg successfully segmented both broad land-cover categories and fine-grained classes in the Sto. Niño dataset.

Profiling analysis demonstrated that VHR-Seg achieves efficient computation in both data pre-processing and model training. Beyond computational efficiency, the specially designed HR-series model achieved competitive performance compared with state-of-the-art (SOTA) methods on natural images. In addition, analyses of ground sampling distance (GSD) and training dataset scale revealed important trade-offs between data quality, dataset size, and model performance, highlighting the adaptability of VHR-Seg to different practical scenarios.

VHR-Seg predictions have applications across different domains, including precision agriculture, biodiversity estimation, and Land Use and Land Cover (LULC) mapping. Additional work is needed to lower computational requirements and to enable the deployment of DL-based solutions on less powerful platforms. Future research may extend VHR-Seg to multiple data

sources such as hyperspectral imagery and LiDAR, or explore domain adaptation strategies to enhance model generalization.

6 Acknowledgements

This dissertation represents not only the culmination of my academic journey but also a deeply personal journey of reconciliation with my own history. I want to start by acknowledging myself for the strength and resilience it took to face and come to terms with the parts of my past that are intricately connected to this research topic. This process has been challenging, yet profoundly meaningful, and I am grateful for the courage I found within myself.

I would like to thank my family and friends, whose love and companionship have carried me through the highs and lows of this process. Your loving presence in my life provided the emotional foundation I needed to embark on this journey of self-reconciliation. Thank you.

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A Application of Segment Anything Model (SAM) and DINOv2 to a section of the Sto. Niño dataset

DINOv2 [32] is a visual foundation model trained via self-supervised learning and capable of extracting rich, general-purpose visual features from unlabeled data. SAM [24] is a zero-shot segmentation model that generates masks for arbitrary prompts at interactive speed. Meta has released an online demo, enabling users to segment objects across a large collection of images. A section of the Sto. Niño dataset was used for this comparison. DINOv2 and SAM cannot segment all banana trees and failed to detect palm trees. However, our VHR-Seg successfully segmented all classes of interest. Note that the image is part of the test set; it was not used during model training.

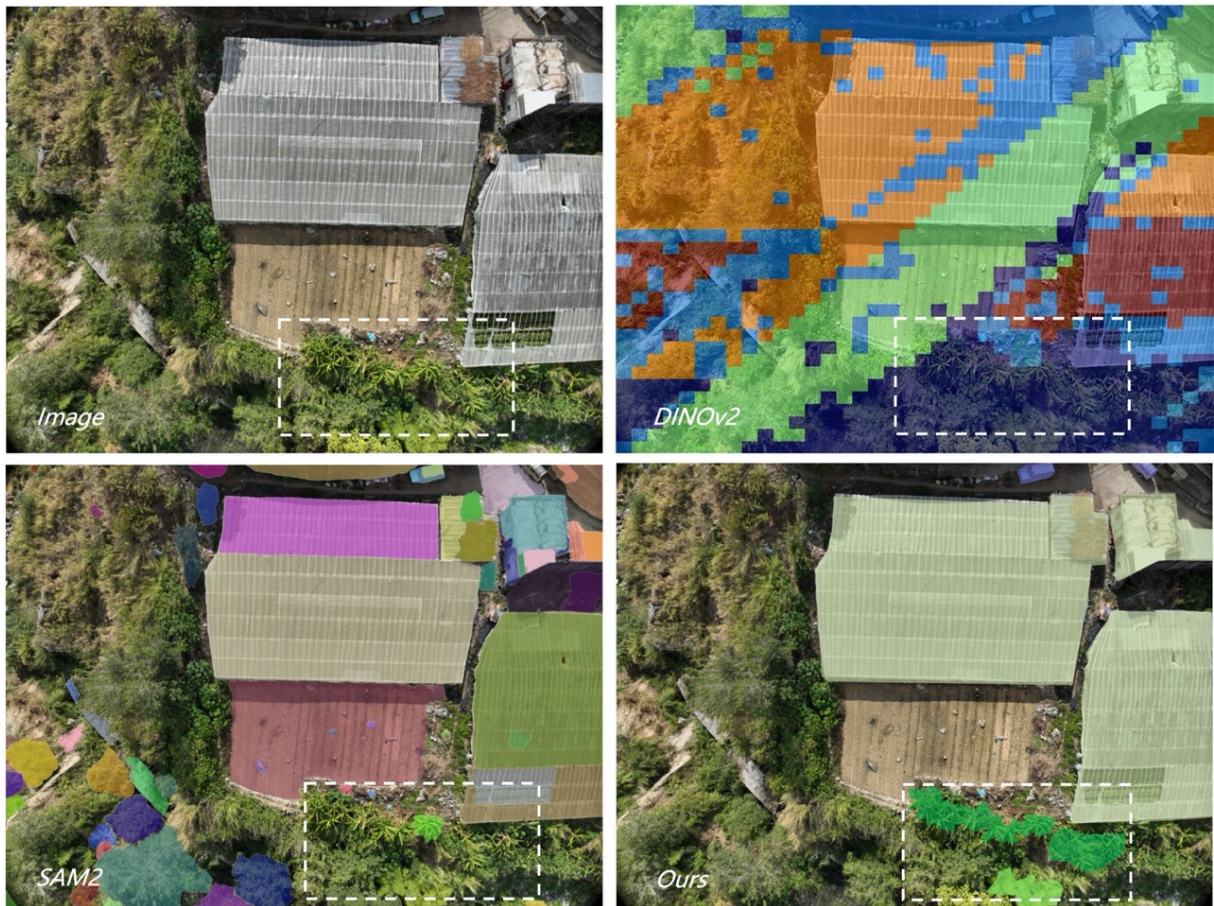


Figure A.1: Application of SAM and DINOv2 to a scene containing banana trees, palm trees, buildings, and cars from a test portion of the Sto. Niño dataset. Clockwise from top left: (1) input image; (2) DINOv2 segmentation output clustered into 5 classes; (3) SAM2 segmentation in automatic mode; (4) segmentation results from our proposed coarse-to-fine system, which successfully identified all target classes.

B Sto. Niño dataset Overview

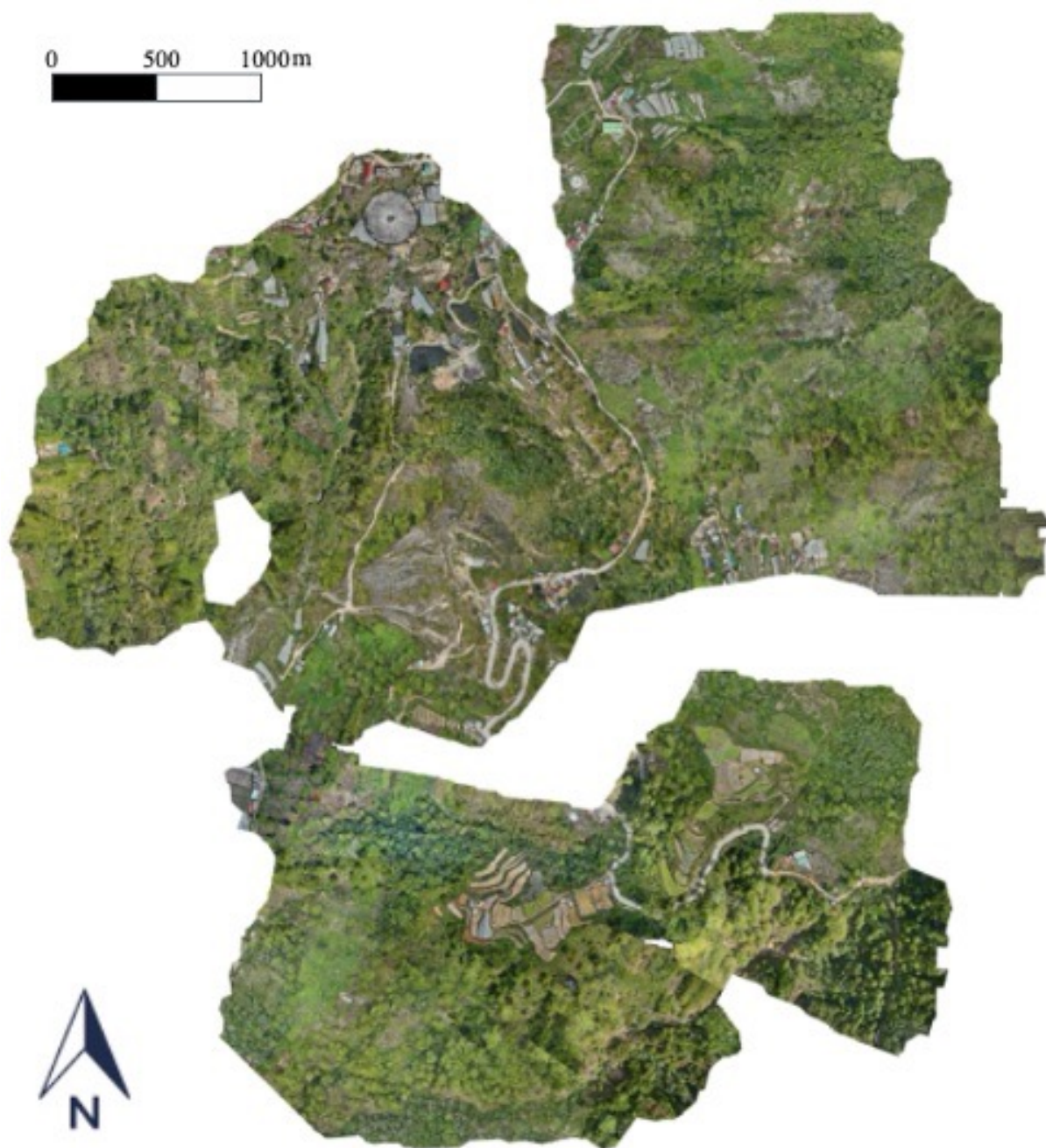


Figure B.1: An overview of the Sto. Niño dataset. The dataset is characterized by its diverse range of vegetation, infrastructure, and varying image quality. The native resolution of around 2cm/px remains fixed over the entire dataset.

C Labeling tutorials for QGIS and CVAT

To support reproducibility, the labeling tutorial documentation provides detailed guidance on the annotation process in QGIS¹. It first gives instructions on installing the required software (QGIS, CVAT, and Docker), followed by step-by-step guidance on importing the dataset and creating image pyramids for efficient navigation. The use of various labeling tools is then demonstrated, and the export process is described to generate segmentation maps in GeoTIFF format. In addition, an alternative tutorial for the CVAT labeling tool is provided, which partially automates label creation. The `label.md` file explains how to generate image tiles with VHR-Seg, upload them to CVAT for labeling, and finally import the generated labels back into VHR-Seg.

D View of Banana trees, Palm trees, and Cars in the Sto. Niño dataset

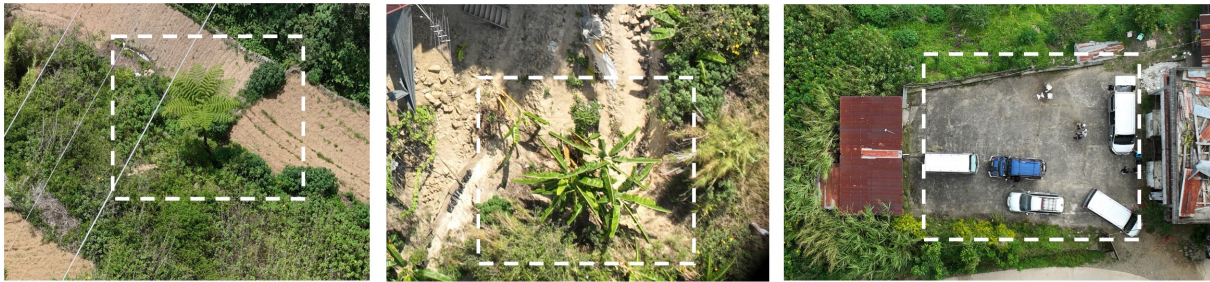


Figure D.1: A partial view of banana trees, palm trees, and cars in the Sto. Niño dataset. The objects of interest are highlighted by dashed boxes for clarity.

E VHR-Seg Task Network

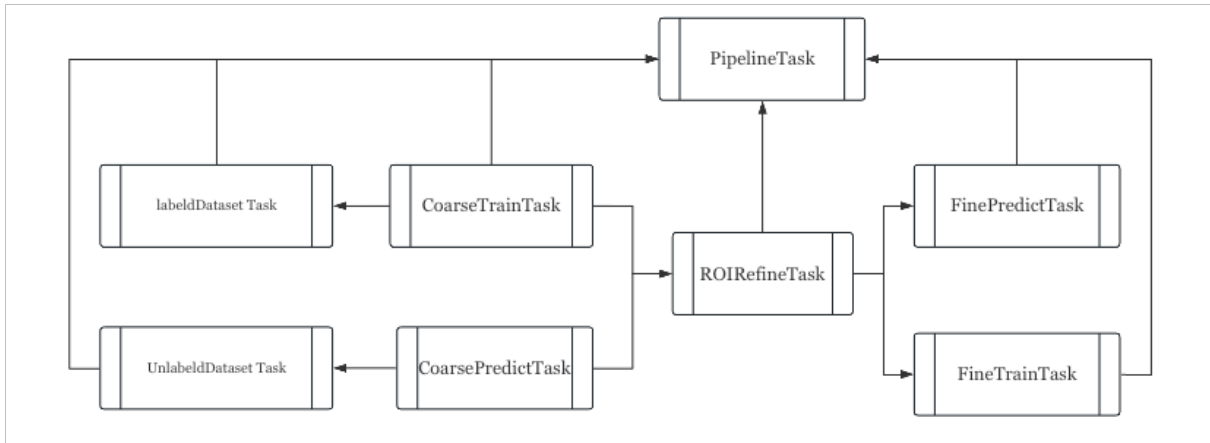


Figure E.1: The VHR-Seg network of Tasks, using UML notation. Arrows indicate dependencies.

¹<https://github.com/ese-ada-lovelace-2024/irp-yc5224>

F Notable features of VHR-Seg

F.1 Parallelized processing

Data processing in VHR-Seg was parallelized using Python's `concurrent.futures` module. The contained `ProcessPoolExecutor` creates a process for each job to run, instead of using threads. VHR-Seg adopts a data pre-chunking strategy, which facilitates effective workload balancing across all processing cores.

F.2 Protection and Hardware compatibility

The long durations tasks may be interrupted due to unexpected resource constraints. VHR-Seg atomically commits the progress of Tasks to a log file, which will be read when restarting. This allows the pipeline to continue computation from the last stable state. VHR-Seg interfaces with the PyTorch training backend to allow training on a wide range of hardware configurations. The code has been tested on CPU-only systems, consumer-grade systems with a single GPU, as well as HPC environments with up to 8 GPUs. Computational requirements for training are adjustable by selecting a particular model variant during model selection, providing trade-offs between training cost and prediction quality.

F.3 Supported data formats and Efficient labeling

The image and label map inputs to VHR-Seg can have any common format, and in particular geospatial formats such as GeoTiff for images and GeoJSON and Shapefiles for label maps are supported. VHR-Seg also supports reading annotations generated with CVAT to help accelerate the labeling procedure.

By employing a coarse-to-fine two-stage segmentation framework, VHR-Seg is able to correct erroneous ground-truth annotations during the coarse stage. Such errors often arise due to differences in annotators' experience, especially in very high-resolution (VHR) RSIs where rich and subtle details increase the likelihood of mislabeling. The hierarchical segmentation design of VHR-Seg provides annotators with the opportunity to revise labels at each stage, thereby reducing error accumulation and preventing the model from learning incorrect feature representations. Moreover, the user interest-guided prediction strategy allows annotators to omit labeling regions that are not of interest, while the region-specific fine model further enhances the accuracy of fine-grained segmentation. And this process decreases the need for labeling.

G Strategies for processing geospatial data in a DL context

G.1 Calculating sample weights for training

Algorithm 1 Calculation of sample weights for the train dataset

Input: Train dataset label tiles $train_dataset_label_tiles$ **Output:** Sample weights $sample_weights$

```
1 Initialize an empty list of lists  $class\_occurrences$ ;  
2 for each label tile  $label\_tile \in train\_dataset\_label\_tiles$  do  
3   Load  $label\_tile$  as grayscale image  $m$  (pixel value = class index);  
4   Count occurrences of each class in  $m$  and append to  $class\_occurrences$ ;  
5 end  
6 Compute  $total\_class\_occurrences$  as the sum of lists in  $class\_occurrences$ ;  
7 Initialize empty list  $sample\_weights$ ;  
8 for each  $sample\_class\_occurrences \in class\_occurrences$  do  
9   Initialize empty list  $s$ ;  
10  for each class  $i$  with occurrence  $class\_occurrence$  do  
11    Append  $\frac{class\_occurrence}{total\_class\_occurrences[i]}$  to  $s$ ;  
12    // Proportion of class  $i$  in this tile relative to dataset;  
13  end  
14  Append  $\sum s$  to  $sample\_weights$ ;  
15 end  
16 return  $sample\_weights$ .
```

G.2 Merging predictions

In the VHR-Seg pipeline, the whole area image is divided into different tiles, which are used as inputs to train or test the model performance. After inference, each image tile has an associated logit tile, where the logits are per-pixel raw scores indicating the model's confidence before normalization (softmax). However, these predictions cannot be directly stitched together into a segmentation map due to overlapping regions between tiles. To address this problem, Researcher [11] developed a merging prediction method, summarized in Algorithm 2.

Algorithm 2 Merging predictions from overlapping tiles

Input: Set of logit tiles L overlapping at pixel p ; Corresponding image tiles T ; Class set C **Output:** Predicted class c_p for pixel p

```
1 for each class  $c \in C$  do  
2   Compute  $m_c = \max_{t \in T} l_t[p][c]$   
3   // Maximum confidence of class  $c$  across all overlapping tiles  
4 end  
5 Set  $c_p = \arg \max_{c \in C} m_c$   
6 return  $c_p$ .
```

H Overview of resolutions

Table H.1: Sto. Niño dataset resolutions and matching platforms

GSD	Resolution (W)	Resolution (H)	Platform
0.08	23662	25228	UAV
0.10	18930	20182	UAV
0.12	15775	16819	UAV
0.15	12620	13455	Maxar (Upscaled)
0.30	6310	6727	Maxar, Digital Globe
0.50	3786	4036	Maxar, Airbus
0.70	2704	2883	Maxar, Planet, CNES, KARI
1	1893	2018	Lockheed Martin Space
3	631	673	Planet
5	379	404	RapidEye Blackbridge
10	189	202	Sentinel-2
15	126	135	Landsat

I Detailed discussion of degradation methods

The degradation methods were illustrated in Figure 3. In Method A, both images and their corresponding labels are downsampled before training. This reduces the amount of information input to the model, but maintaining pixel density. Therefore each pixel represents a larger physical distance. The requirements of different model architecture in this process to accommodate for the new input size, making comparisons with baseline model more difficult. Downsampling of labels reduces the precision of gradient computations when updating model weights, which impedes training progress at low resolutions. `INTER_AREA` interpolation of the OpenCV library [2] was used for downsampling.

Method B (Pixelation) does not affect input size, but also reduces the amount of conveyed information, while lowering pixel density. Unlike method A, labels are not degraded, which gives a better estimate of how image quality alone affects predictions. Method B represents a synthetic study of resolution degradation, as multiple pixels in the final tile represent a single pixel in the downsized tile. `INTER_AREA` interpolation was again used for downsampling, followed by bicubic upscaling.

Method C generates "zoomed-out" tiles compared to the original input [2]. Although pixel-level detail is reduced, the model sees a broader spatial context. Pixel density has not changed, so each pixel represents a larger ground area. This simulates resolution changes under higher altitude or lower-resolution sensors. Lower orthomosaic resolution results in fewer training tiles, which may lead to overfitting. All degraded datasets were produced via Lanczos downsampling in QGIS [33].

J HR-series Model design

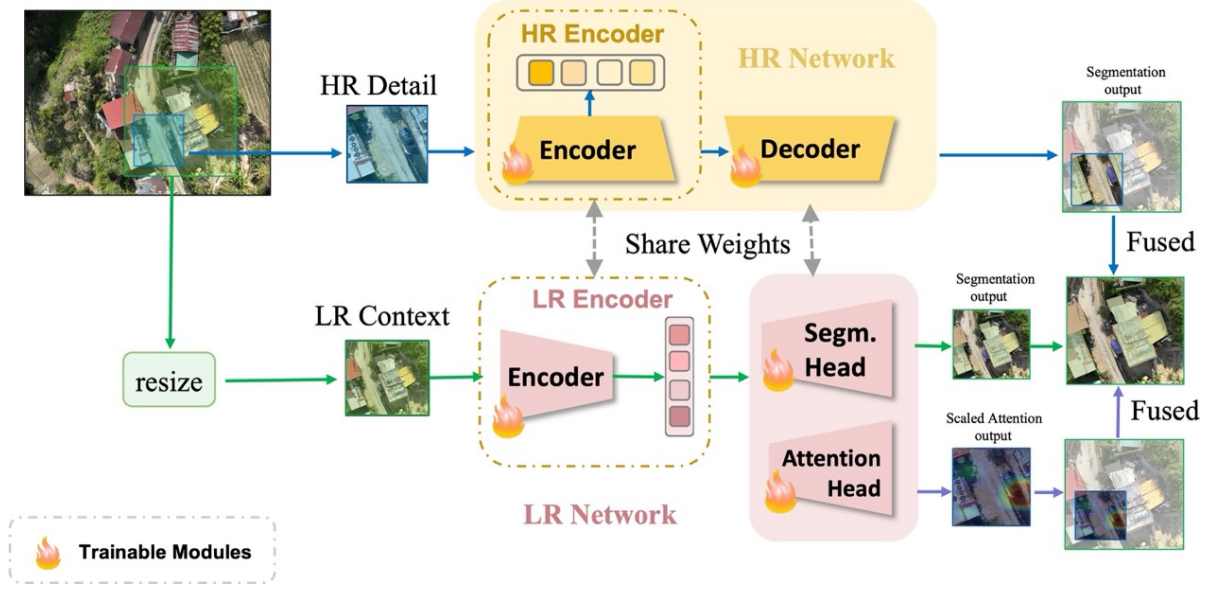


Figure J.1: The HR-series architecture of the proposed model and datasets.

In this section, we describe the HR-series model, which acts as an encoder-decoder module. It is wrapped around the original encoder-decoder backbones of Segformer and Mask2former, resulting in two extended models: HRsegformer and HRmask2former.

HR-series model preliminary. To better utilize the advantages of very high resolution of data, HR model employs high-resolution (HR) crops to capture detailed features while simultaneously using low-resolution (LR) crops to preserve global context. Define x^t is one tile in orthophoto or one image from UAV raw data. $\mathcal{F}$ represents the segmentation model.

LR and HR crop. The combination of multi-level features enables the model to not only focus on the detail features, but also allow the model to make sure it will be gotten in a right context. The context crop x_t^{LR} is cropped from the original HR resolution target domain image x_t^{HR} and bilinear down sampling by the factor s .

$$x_t^{LR} = \zeta(\text{Crop}(x_t^{HR}, \mathbf{b}_L), 1/s), \quad (3)$$

where Crop stands for crop operation, $\mathbf{b}_L = (b_{L,1}, b_{L,2}, b_{L,3}, b_{L,4})$ represents the cropping area, ζ is down sampling operation. We sample the cropping area $\mathbf{b}_{LR}$ in three steps. First, compute the grid interval $k = s \cdot o$, where $o \geq 1$ is the output stride of the segmentation network. Then partition the row and column indices of the high-resolution image into two regular grids with step k ; only these grid points are regarded as valid top-left corners for a crop. From each grid, draw one index with a discrete uniform distribution to obtain the vertical and horizontal start coordinates $(b_{L,1}, b_{L,3})$. To keep the crop inside the image and to fix its size, set the bottom-right corner to $b_{L,2} = b_{L,1} + s h_L$ and $b_{L,4} = b_{L,3} + s w_L$. Consequently, all four edges of the crop fall exactly on the feature-map grid, which guarantees pixel-level correspondence during the subsequent multi-scale fusion stage. The detail crop area is randomly cropped from context crop:

$$x_{t,LR}^{HR} = \text{Crop}(x_t^{LR}, \mathbf{b}_H), \quad (4)$$

where $\mathbf{b}_H = (b_{H,1}, b_{H,2}, b_{H,3}, b_{H,4})$ with a size of $h_H \times w_H$ represents the detail cropping area. And it is randomly cropped from $\mathbf{b}_L$. The context semantic segmentation is $y_t^{LR} = \mathcal{F}(x_t^{LR})$

and detail semantic segmentation is $y_{t,LR}^{\hat{H}R} = \mathcal{F}(x_{t,LR}^{HR})$, $\mathcal{F}$ is the semantic segmentation network.

Multi-Resolution Fusion. To balance fine details with global context, we use a learnable scale-attention mechanism to adaptively fuse predictions from the HR detail crop and LR context crop. A attention decoder F_T^A learns to predict the scale attention $a_{LR} = \sigma(F_T^A(F_T^E(x^{LR})))$ to weigh the trustworthiness of LR context and HR detail predictions. The sigmoid function σ ensures a weight in $[0, 1]$, where 1 means a focus on the HR detail crop. The attention is predicted from the context crop as it has a better grasp on the scene layout (larger context). As the predictions are smaller than the inputs due to the output stride o , the crop coordinates are scaled accordingly in the following steps. Outside of the detail crop, the attention is set to 0 as there is no detail prediction available and get a'_{LR} . The HR detail crop is aligned with the (upsampled) context crop by padding it with zeros to a size of $\frac{sh_L}{o} \times \frac{sw_L}{o}$ and get $y_{LR}^{\hat{H}R'}$. The predictions from multiple scales are fused using the attention-weighted sum:

$$\hat{y}_{LR,F} = \zeta((1 - a'_{LR}) \odot \hat{y}_{LR}, s) + \zeta(a'_{LR}, s) \odot y_{LR}^{\hat{H}R'}. \quad (5)$$

The encoder, segmentation head, and attention head are trained with the fused multi-scale prediction and the detail crop prediction. The target loss $\mathcal{L}_{CRAM}^T$ is:

$$\mathcal{L}_{CRAM}^T = (1 - \lambda_d) \mathcal{L}_{ce}(\hat{y}_{LR,F}^T, p_{LR,F}^T, q_{LR,F}^T) + \lambda_d \mathcal{L}_{ce}(\hat{y}_{HR}^T, p_{HR}^T, q_{HR}^T), \quad (6)$$

where $\lambda_d \in [0, 1]$: balance factor between the low-resolution context branch and the high-resolution detail branch. $\mathcal{L}_{ce}(\cdot, \cdot, \tau)$: temperature-scaled cross-entropy, where the temperature τ sharpens ($\tau < 1$) or smooths ($\tau > 1$) the label distribution. $\hat{y}_{LR,F}^T$ is the logits of the scale-attention-fused context prediction after up-sampling to the original image size. $p_{LR,F}^T$ is label map that aligns spatially with $\hat{y}_{LR,F}^T$. $\hat{y}_{HR}^T$ is logits predicted on the high-resolution detail crop and p_{HR}^T is labels for the detail crop. The hyper parameter s and relative crop size will be analysed in Appendix K.1

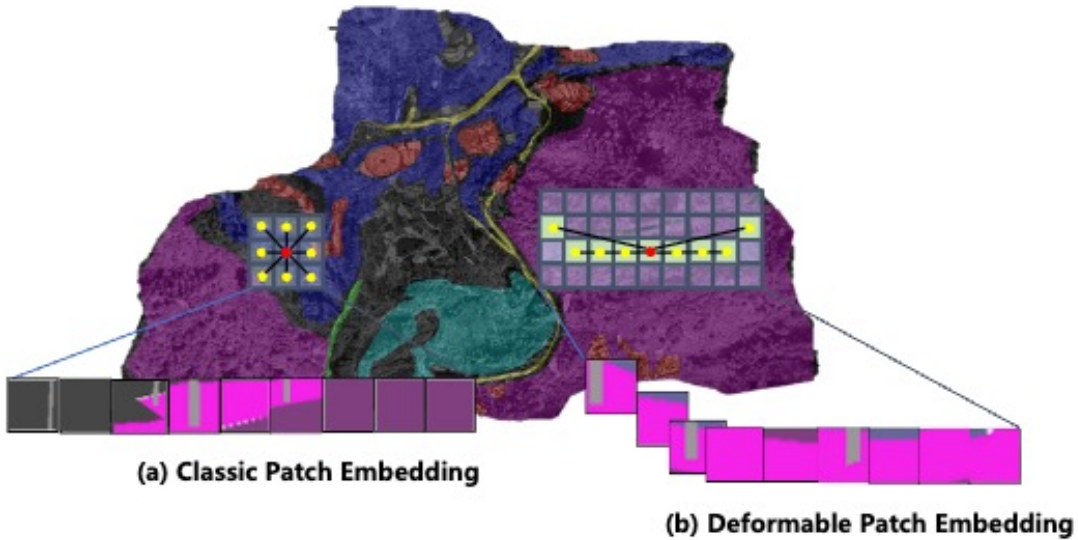


Figure J.2: Visualization of HR-series feature extraction

K Ablation Study

K.1 Analysis of hyper-parameter down sampling factor s and relative crop size

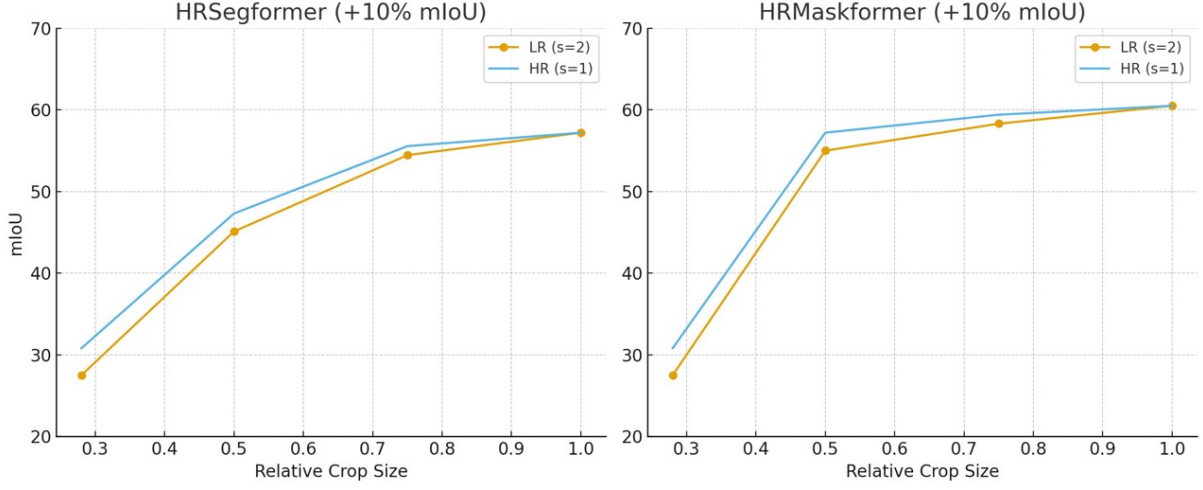


Figure K.1: Segmentation performance over the relative crop size for different resolutions and for both HRsegformer and HRmask2former.

A hyper-parameter analysis focusing on the relative crop size and resolution scale is further conducted. As shown in Figure J.2, both HRsegformer and HRmask2former consistently benefit from larger crops, where the mIoU improves steeply from 0.3 to 0.75 and then saturates between 0.75 and 1.0. This observation is consistent with our fine-grained stage analysis. Small crops in the early stage encourage the learning of global context, medium crops (0.5–0.75) in the mid stage enhance boundary detection and small-object recognition, while larger crops (≥ 0.75) in the late stage mainly stabilize predictions with diminishing returns.

For the downsampling factor s , high-resolution input ($s = 1$) enables faster convergence and more accurate boundary modeling, whereas low-resolution input ($s = 2$) enhances global invariance and alleviates overfitting in the early phase.

The performances of the HR-series models are also different: HRsegformer shows more balanced gains on vegetation and road-like classes, while HRmask2former performs better on structured objects such as cars and buildings. In particular, we find that using $s = 2$ with a relative crop size of 0.5 provides a good trade-off in the fine-grained stage, while larger crops combined with HR inputs are more beneficial in the late stage.

L Numeric data on per-class model performance relative to GSD

Table L.1: Quantitative evaluation of coarse stage segmentation performance as GSD is increased

Method	GSD (m/px)	mIoU	mDice	IoU Background	IoU Building	IoU Road	IoU Forest	Best val. iteration
n/a	0.08	81.39	82.83	99.64	80.35	74.28	71.32	15000
A	0.10	79.23	81.43	99.08	75.64	71.73	70.48	10000
	0.12	77.51	79.48	99.08	72.38	70.26	68.32	5000
	0.15	74.83	77.21	98.34	65.43	68.32	67.24	2500
	0.30	67.13	70.34	97.6	50.32	55.21	65.40	1250
	0.50	57.33	60.12	96.96	35.24	33.85	63.26	625
	0.70	52.15	55.36	95.76	20.15	30.55	62.15	312
	1	48.18	50.88	96.74	15.33	20.23	60.42	128
	3	38.62	43.16	98.97	5.26	0.00	50.24	10
	5	32.57	33.27	99.64	0.00	0.00	30.65	6
	10	28.10	30.32	99.91	0.00	0.00	12.47	5
	15	26.29	38.33	99.96	0.00	0.00	5.21	3
B	0.10	78.88	81.70	98.36	76.42	70.86	69.88	13500
	0.12	77.47	78.09	98.34	73.95	69.80	67.80	15000
	0.15	75.74	76.72	97.84	71.10	67.20	66.80	10000
	0.30	67.29	70.11	97.97	51.45	56.32	63.40	9500
	0.50	57.62	64.18	97.11	40.37	32.54	60.45	13000
	0.70	54.39	59.76	96.52	35.34	30.46	55.25	11500
	1	47.30	52.85	96.15	12.75	23.56	56.75	7500
	3	42.08	49.53	94.59	5.60	17.84	50.28	13500
	5	31.35	40.71	92.34	0.00	0.00	33.05	12500
	10	25.42	29.05	88.19	0.00	0.00	13.47	15500
	15	23.07	19.16	88.83	0.00	0.00	3.43	13500
C	0.10	79.48	80.97	98.46	77.33	71.25	70.86	8500
	0.12	79.05	75.48	98.60	78.24	69.98	69.36	6500
	0.15	76.92	72.81	98.34	72.79	68.32	68.23	2500
	0.30	69.71	71.34	97.60	56.04	58.24	66.95	1528
	0.50	53.56	60.12	96.96	25.43	31.56	60.28	630
	0.70	46.40	49.26	95.76	18.73	13.78	57.34	342
	1	41.00	48.88	96.74	4.57	6.32	56.38	90
	3	36.99	38.16	98.97	0.00	0.00	48.75	10
	5	32.48	33.27	99.64	0.00	0.00	30.26	6
	10	28.37	31.32	99.91	0.00	0.00	13.56	5
	15	26.07	29.33	99.96	0.00	0.00	4.32	3

Table L.2: Quantitative evaluation of coarse stage segmentation performance as train dataset size is decreased

images in Train Set	mIoU	mDice	IoU Road	IoU Forest	IoU Building	IoU Background
3001	79.25	74.83	71.43	70.48	75.64	99.45
1963	75.9	75.81	69.85	64.23	70.38	99.12
1388	71.58	76.37	66.74	55.24	65.32	99.02
910	66.26	73.81	55.62	50.43	60.24	98.75
254	52.40	67.02	50.47	30.26	30.45	98.45
93	55.31	66.03	32.41	25.42	20.25	98.32
50	44.1	48.94	20.56	18.31	15.31	98.56
23	30.71	34.93	20.35	4.28	0.00	98.21
2	24.55	31.66	0.00	0.00	0.00	98.21
1	24.55	32.39	0.00	0.00	0.00	98.21

Table L.3: Quantitative evaluation of fine-grained stage segmentation performance as train dataset size is decreased

images in Train Set	mIoU	mDice	IoU Background	IoU Banana tree	IoU Palm tree
3001	71.54	74.83	98.64	60.25	55.73
1963	68.56	75.81	98.65	53.21	53.76
1388	60.53	76.37	98.71	40.32	42.56
910	55.10	73.81	98.36	35.44	31.49
254	48.44	67.02	96.93	28.53	19.85
93	12.69	66.03	96.86	17.32	0.00
50	33.59	48.94	95.31	5.46	0.00
23	30.87	34.93	92.61	0.00	0.00
2	28.07	31.66	84.21	0.00	0.00
1	30.85	32.39	92.55	0.00	0.00

Table L.4: Quantitative evaluation of fine-grained segmentation performance as GSD is increased

Method	GSD (m/px)	mIoU	mDice	IoU Background	IoU Banana tree	IoU Palm tree	Best val. iteration
n/a	0.08	71.87	73.83	99.64	60.25	55.73	15000
A	0.10	71.48	73.97	98.46	60.25	55.73	10000
	0.12	69.99	73.48	98.60	59.84	51.54	5000
	0.15	66.40	72.81	98.34	55.32	45.53	2500
	0.30	62.75	66.34	97.6	50.33	40.32	1250
	0.50	56.91	60.12	96.96	42.49	31.28	625
	0.70	43.66	49.26	95.76	29.99	5.24	312
	1	38.93	48.88	96.74	18.62	1.45	128
	3	32.99	33.16	98.97	0.00	0.00	10
	5	33.21	33.27	99.64	0.00	0.00	6
	10	33.3	33.32	99.91	0.00	0.00	5
	15	33.32	33.33	99.96	0.00	0.00	3
B	0.10	70.07	72.68	98.36	58.32	53.54	13500
	0.12	69.91	73.09	98.34	59.84	51.54	15000
	0.15	67.18	76.72	97.84	56.32	47.38	10000
	0.30	64.58	70.11	97.97	51.45	44.32	9500
	0.50	56.67	62.18	97.11	40.37	32.53	13000
	0.70	54.11	59.76	96.52	35.34	30.46	11500
	1	35.42	42.85	96.15	5.75	4.36	7500
	3	31.53	33.67	94.59	0.00	0.00	13500
	5	30.78	36.71	92.34	0.00	0.00	12500
	10	29.40	33.05	88.19	0.00	0.00	15500
	15	29.61	34.16	88.83	0.00	0.00	13500
C	0.10	69.62	76.97	98.46	59.84	50.56	8500
	0.12	68.72	73.48	98.60	58.24	49.32	6500
	0.15	66.22	72.81	98.34	52.79	47.53	2500
	0.30	61.93	66.34	97.60	46.04	42.15	1528
	0.50	51.32	60.12	96.96	25.43	31.56	630
	0.70	42.09	49.26	95.76	16.73	13.78	342
	1	37.21	43.88	96.74	8.57	6.32	90
	3	32.99	38.16	98.97	0.00	0.00	10
	5	33.21	33.27	99.64	0.00	0.00	6
	10	33.30	31.32	99.91	0.00	0.00	5
	15	33.32	30.33	99.96	0.00	0.00	3

M Visualization model performance with different survey characteristics

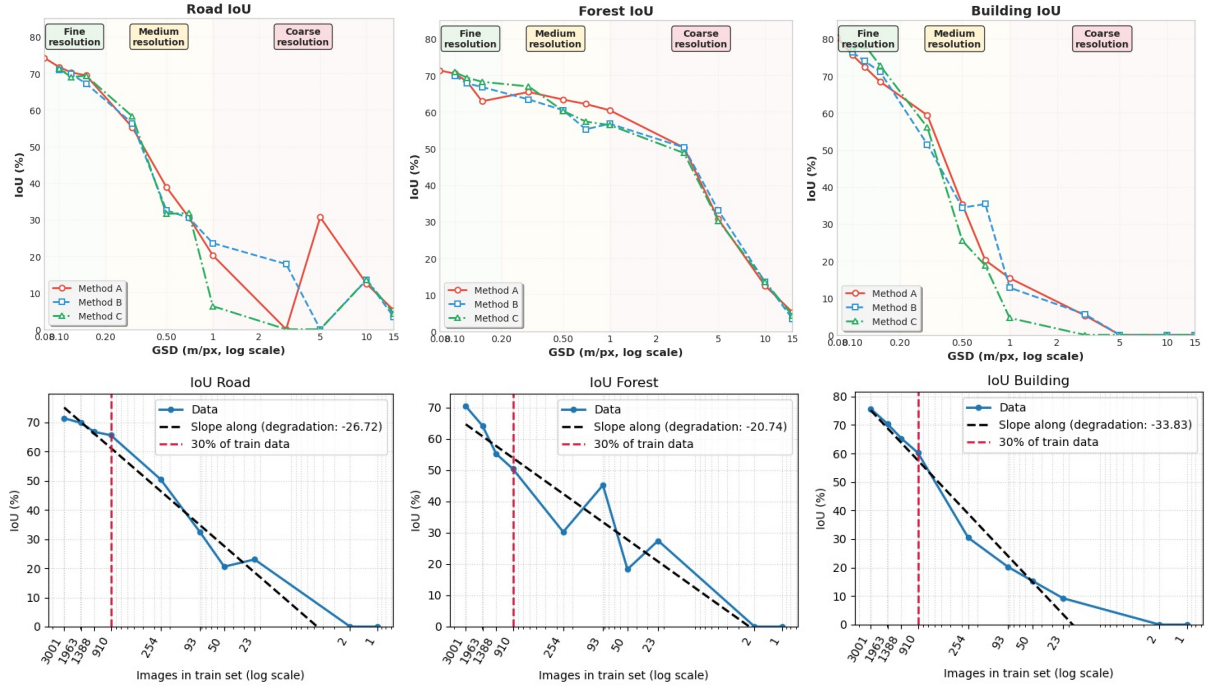


Figure M.1: Model performance visualization under varying survey characteristics in coarse stage. Top row shows the IoU trends for Road, Forest, and Building classes across different sampling scales, while the bottom row presents detailed comparisons of IoU versus the number of training samples.

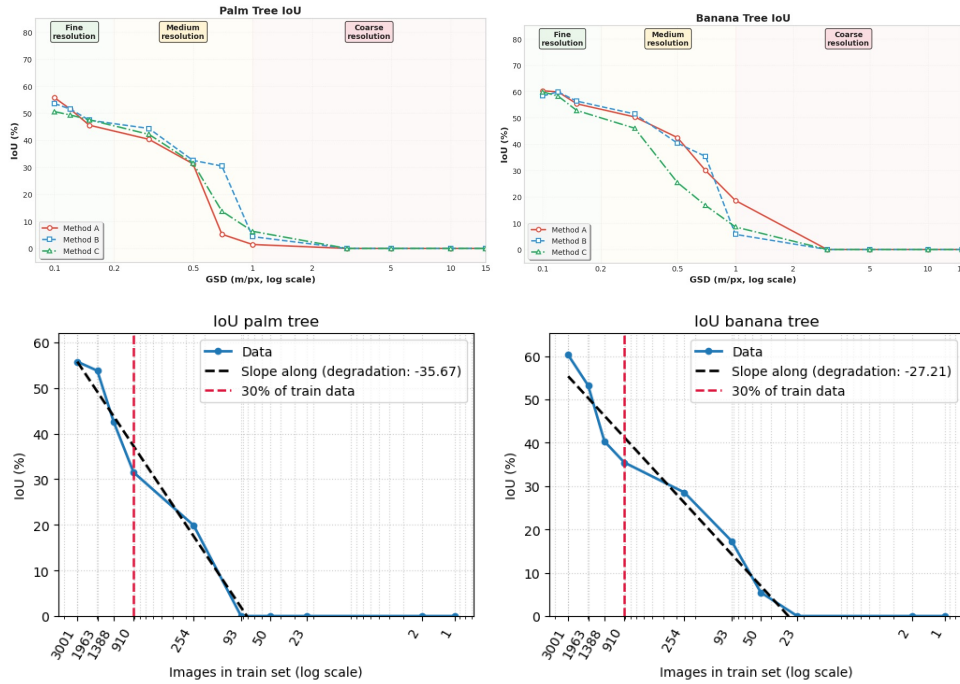


Figure M.2: Model performance on fine-grained vegetation classes in fine-grained stage. Top row shows IoU degradation trends for Palm Tree and Banana Tree under different methods, while bottom row presents IoU versus training set size with slope degradation analysis.

N Variation of Survey Extent with Spatial Resolution

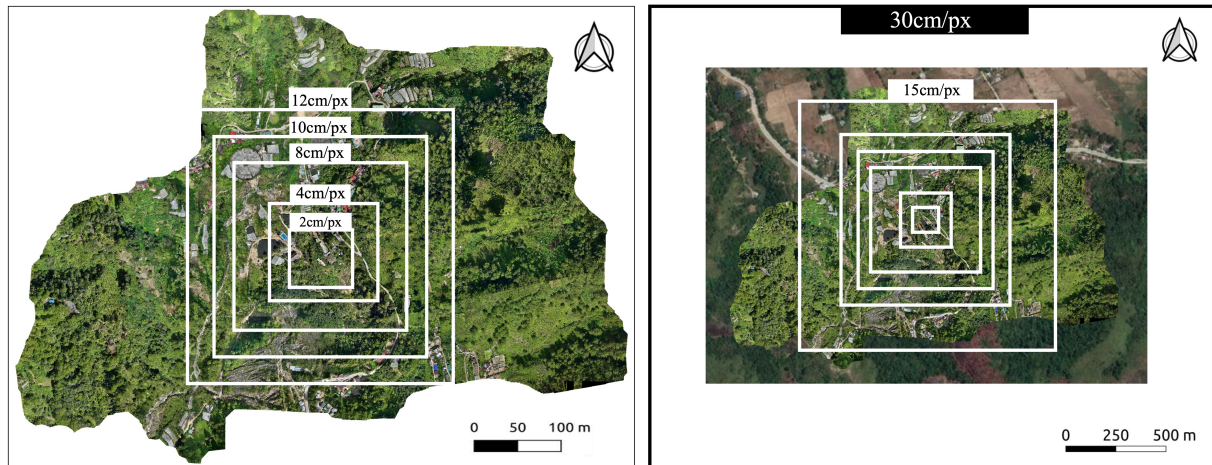


Figure N.1: Variation of survey extent with spatial resolution. Left panel: Sto. Niño dataset overlaid onto satellite imagery, with bounding boxes representing different very high resolution (VHR) ground sampling distances (2–12 cm/px). Right panel: additional extents corresponding to coarser resolutions (15–30 cm/px).

Table N.1: Estimated flight statistics for a DJI Mavic 3M drone imaging each region shown in Fig. N.1, such that the number of pixels in each orthomosaic is the same as for the native dataset with a GSD of 0.022 m/px. Statistics for GSDs of 3, 5, 10, and 15 m/px not shown as the maximum flight altitude of 6km would be exceeded. The maximum legal flight height in most countries is 120 metres, thus a lower quality sensor would need to be fitted on the drone, rather than increasing altitude as shown here. Calculations are based on the specifications provided by DJI.

GSD (m/px)	Altitude (m)	Area to image (km ²)	8h workdays to complete imaging (optimistic estimates)
0.022	75.08	3.06	0.30
0.04	150.16	10.12	1.34
0.08	300.32	40.48	2.22
0.10	375.40	63.26	4.44
0.12	450.48	91.09	5.55
0.15	563.098	142.32	6.66
0.30	1126.20	569.30	8.33
0.50	1876.99	1581.38	16.66
0.70	2627.79	3099.50	27.76
1.00	3753.99	6325.51	38.87